

Likelihood with multiple data points:

$$\vec{d} = (d^0, d^1, \dots, d^M)$$

$$\vec{n} = (n^0, n^1, \dots, n^M)$$

$$\vec{r} = \dots$$

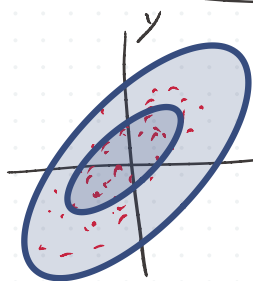
$$\vec{h} = \dots$$

$$p(d|h) \sim \mathcal{N}(0, C)$$

$$C_{ij} = \langle n^i n^j \rangle, \quad C_{ij} \text{ } M \times M \text{ symmetric}$$

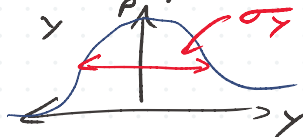
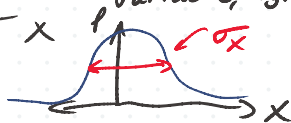
$$p(d|h) = \frac{1}{(2\pi)^{M/2}} \frac{1}{|C|^{1/2}} \cdot e^{-\frac{1}{2} C^{-1}_{ij} r^i r^j}$$

Gaussian Likelihood



Joint probability distribution (Joint PDF)

Marginal distributions (pick one variable, ignore the rest).

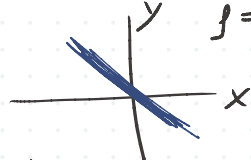


$$C_{ij} = \begin{pmatrix} \langle xx \rangle & \langle xy \rangle \\ \langle xy \rangle & \langle yy \rangle \end{pmatrix} = \begin{pmatrix} \sigma_x^2 & \rho \sigma_x \sigma_y \\ \rho \sigma_x \sigma_y & \sigma_y^2 \end{pmatrix}$$

Cross-correlation
How much is the noise correlated?
 $\rho \in [-1, 1]$

Auto correlation
"How much noise is in the data?"

Perfect anti-correlation
 $\rho = -1$



$$\det C_{ij} = 0$$

Zero correlation
 $\rho = 0$



$$C_{ij} = \sigma_i^2 \cdot \delta_{ij} \\ \det C = \prod_i \sigma_i^2$$

Perfect correlation
 $\rho = +1$



$$\det C_{ij} = 0$$

$\vec{r}^T \cdot \vec{C}^{-1} \cdot \vec{r} \rightarrow O(N^2)$ operations ($O(N^3)$ if you calculate C^{-1})

Stochastic part of data

- noise
- background sources

Unpredictable, we don't have a model for the shape $\Rightarrow$ Model variance only.



Discrete part of data

- GW events with a waveform
- Transient noise events

LVLK "glitches"



Fit with model and subtract it.

C_{ij} is symmetric, so $C_{ij} = C_{ji}$

$M(M+1)/2$ degrees of freedom.

Time domain data:

$$\vec{r} = [r_1, r_2, \dots, r_N]$$

$$C_{ij} = \langle r_i r_j \rangle = \langle r(t_i) r(t_j) \rangle \equiv R(t, \tau)$$

$$\tau = t_j - t_i \quad \text{"noise"}$$

If $C_{ij} = \sigma^2 \delta_{ij} \Rightarrow$ "white noise". autocorrelation

If $R(t, \tau) = R(\tau) \Rightarrow$ "stationary noise"

$$R(\tau) = \langle r(t) r(t+\tau) \rangle$$

In that case C_{ij} has a special form:

$$C_{ij} = C_{i+k, j+k} \equiv C_k$$

$$C_{ij} = \begin{pmatrix} C_0 & C_1 & C_2 & \dots & \dots \\ C_1 & C_0 & C_1 & C_2 & \dots \\ C_2 & C_1 & C_0 & C_1 & C_2 & \dots \\ \vdots & \vdots & \vdots & \ddots & \vdots & \ddots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Toeplitz Matrix

Only has N degrees of freedom!

We can diagonalize C_{ij}

$$U \cdot C \cdot U^{-1} = \tilde{C} \text{ (diagonal)}$$

$$\vec{r} = U \cdot \vec{r}$$

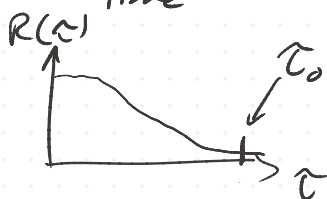
$$p(\vec{r}) = \mathcal{N}(0, \tilde{C}^{-1}) \rightarrow p(r) \sim \mathcal{N}(0, \sigma^2)$$

U : unitary matrix
 $\det U = 1$

In the limit where $T \gg \tau_0$

Observation time

"Auto correlation time"



$$U_{jk} \sim N^{-1/2} e^{2\pi i j k N}$$

Discrete Fourier transform!

In this case,

$$p(d|h) = \frac{1}{(2\pi)^{N/2}} \frac{1}{|C|^{1/2}} e^{-\vec{r}^T \cdot C^{-1} \cdot \vec{r}}$$

(Same formula, but in Fourier domain).

(Next class, Fourier transform refresher).

2x2 example

$$C_{ij} = \begin{pmatrix} C_{11} & C_{12} \\ C_{12} & C_{22} \end{pmatrix} \Rightarrow U_{ij} = \begin{pmatrix} e^{i\pi} & e^{2\pi i} \\ e^{2\pi i} & e^{4\pi i} \end{pmatrix} \cdot \frac{1}{\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$U_{ij}^{-1} = -\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

$$C \cdot U^T = \frac{-1}{\sqrt{2}} \begin{pmatrix} C_{11} & C_{12} \\ C_{12} & C_{22} \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

$$= \frac{-1}{\sqrt{2}} \begin{pmatrix} C_{11} - C_{12} & -C_{11} - C_{12} \\ C_{12} - C_{22} & -C_{12} - C_{22} \end{pmatrix}$$

$$U \cdot C \cdot U^T = \frac{-1}{2} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} C_{11} - C_{12} & -C_{11} - C_{12} \\ C_{12} - C_{22} & -C_{12} - C_{22} \end{pmatrix}$$

$$= \frac{-1}{2} \begin{pmatrix} -(C_{11} - C_{12}) + C_{12} - C_{22} & C_{11} + C_{12} - C_{12} - C_{22} \\ C_{11} - C_{12} + C_{12} - C_{22} & -C_{11} - C_{12} - C_{12} - C_{22} \end{pmatrix}$$

$$\tilde{C}_{11} = \frac{C_{11} + C_{22}}{2} + C_{12} \quad (\text{check math})$$

$$\tilde{C}_{12} = 0 \quad (\text{should be})$$

$$\tilde{C}_{22} = \frac{C_{11} + C_{22}}{2} + C_{12}$$

$$\Rightarrow \tilde{C} = \langle \tilde{n}(f) \tilde{n}^*(f') \rangle = \langle n^2(f) \rangle \delta(f - f')$$

"power spectrum"

$$\langle n^2(f) \rangle = \int \underbrace{\langle n(t) n(t - \tau) \rangle}_{n \sim \sqrt{T} \text{ (random walk)}} e^{i\pi f \tau} d\tau$$

$n \sim \sqrt{T}$ (random walk)

$$\langle n^2(f) \rangle \sim T$$

$$Z_{ij} = \langle \tilde{x}(f_i) \tilde{x}^*(f_j) \rangle = \left(\frac{T}{2} \right) S_n(f_i) \delta(f_i - f_j)$$

T b/c $\langle n^2 \rangle \sim T$

Z b/c real, imag

parts of $\tilde{x}(f)$

Power Spectral density

PSD

$$p(\vec{I} | \vec{h}) = \frac{1}{\sqrt{T!} \prod_i \pi T S_{n,i}} e^{-\frac{1}{2} \sum_i \frac{\tilde{r}_i^* \tilde{r}_i}{\frac{T}{2} S_{n,i}}} \rightarrow \mathcal{O}(N)$$

Fourier domain

Gaussian likelihood